JÖNKÖPING UNIVERSITY
Jönköping International
Business School

Demand volatility in logistics operations and its effects on operational management within parcel service providers

Bachelor Degree Project

Thesis Within: Business Administration

Number of Credits: 15hp

Programme of Study: International Management

Authors: Arseni Kazlouski, Kevin De Silva

JÖNKÖPING May 2026

Abstract

The recent explosion in popularity of e-commerce has contributed to increased volatility of demand in parcel logistics chains, giving rise to the conflict between the rigidity of the physical and human resources and unpredictable volumes of parcels generated by consumers' orders. While there is a considerable body of research dedicated to last-mile logistics from a strategic point of view, surprisingly little research has focused on the impact of this phenomenon on the operations of the people involved. This paper explores how demand volatility affects the operational activities of parcel service companies, focusing on how volume variations affect route planning and capacity use, what problems emerge when dealing with high and low demands, what adjustment methods are used, and whether they lead to solving the initial problem or to secondary issues.

This paper employs qualitative design of case studies based on semi-structured interviews with operational managers, terminal workers, coordinators, and drivers of five parcel delivery organizations – UPS, Airmee, Bring, Bonway and PostNord – in Sweden, including representatives from different organizational levels, from drivers to area managers. Thematic analysis with six pre-established themes was performed to analyze the results.

As a result, it was found that demand volatility occurs systematically and depends on the peculiarities of each organization, following company-specific weekly and seasonal patterns caused mainly by e-commerce. High volumes pose capacity constraints for sorting operations, human resources, and transportation vehicles, while low volumes pose idle cost concerns due to similar constraints. The coping strategies include flexible scheduling, real-time route adjustment, and managing capacity outside the organization; however, all of them have their own secondary costs. Information asymmetry, uncertainty associated with upstream supply chain dynamics, and lack of real-time destination information emerged as the reasons why all organizations struggled to implement plans in practice. Overall, the main finding was that demand volatility was handled but not solved, with experiential judgment playing an essential role alongside technologies-based forecasting techniques.

Table of Content

Abstract.....1

1 Introduction.....4

1.1 Problem Statement.....4

1.2 Purpose of the Study.....5

1.3 Research Questions.....5

2 Literature Review/ Theoretical framework.....6

2.1 Literature Review Methodology.....6

2.2 Introduction to the Theoretical Framework.....7

2.3 Parcel Service Providers: Operational Networks and Structural Challenges.....8

2.4 Demand Variability and Volatility in parcel service providers.....9

2.5 Operational Challenges Arising from Demand Volatility.....10

2.6 Managerial Responses and Adjustment Strategies.....11

2.7 Decision-Making Under Uncertainty.....13

2.8 Planning vs Execution: The Operational Gap.....13

2.9 Organizational Learning and Adaptation.....14

3 Methodology.....15

3.1 Research Design.....15

3.2 Data Collection.....15

3.2.1 Primary Data.....15

3.2.2 Secondary Data.....17

3.3 Sampling Strategy.....17

3.4 Interview Guide.....18

3.5 Data Analysis Approach.....18

3.6 Research Quality and Limitations.....19

3.7 Ethical Considerations.....19

4 Findings.....20

4.1 Overview of Participating Organizations and Respondents.....20

4.2 Experience of Demand Volatility.....21

4.2.1 Seasonal Patterns.....21

4.2.2 Intra-Weekly Patterns.....22

4.2.3 Drivers of Fluctuation.....22

4.3 Operational Challenges During High-Demand Periods.....23

4.4 Challenges Arising during Low Demand Times.....25

4.5 Managerial Responses and Adjustment Strategies.....26

4.6 Decision-Making Under Uncertainty.....27

4.7 Planning-Level Versus Execution-Level Perspectives.....29

4.8 Communication Structure and Feedback Systems.....30

4.9 Organizational Learning and Future Adaptation.....31

5. Analysis.....33
5.1 Experience of Demand Volatility and Its Drivers..... 34
5.2 Operational Challenges..... 37
5.3 Managerial Responses and Adjustment Strategies.....39
5.4 Decision-Making Under Uncertainty.....40
5.5 Planning-Level vs Execution-Level Perspectives.....42
5.6 Organizational Learning and Future Adaptations.....43
5.7 Synthesis of Findings..... 44
5.8 Conclusion of Analysis..... 45
6 Conclusions.....46
Fluctuations in parcel volume: What impact do they have on route planning, stop density, and resource utilization throughout various operational shifts?.....46
High-demand periods vs low-demand periods: What operational challenges do they pose and what are the differences?..... 46
What strategies do managers and coordinators use to tackle demand variability?..... 47
Do the strategies described alleviate the strain, or do they create additional inefficiencies? 47
7 Discussion.....48
7.1 Beyond the Empirical Findings.....48
7.2 Implications for Practitioners.....49
7.3 Implications for Future Research.....49
7.4 Methodological Reflection.....50
References.....51
Appendix A: Interview Guide.....55
Introduction for Participants and consent.....55
Theme 1: Experience of Demand Volatility..... 55
Theme 2: Operational Challenges.....56
Theme 3: Managerial Responses and Adjustment Strategies..... 56
Theme 4: Decision-Making Under Uncertainty.....56
Theme 5: Planning-Level vs Execution-Level Perspectives.....56
Theme 6: Organizational Learning and Future Adaptation..... 56

1 Introduction

During the past decade, e-commerce has rapidly developed and has impacted the parcel logistics industry greatly. Over the past two decades, the rate of online purchases increased globally from 1.7% in 2003 to 20% in the post-pandemic era, indicating changes in customers' buying behavior (Boyer & Hult, 2005; World Economic Forum, 2025). In addition, order batching was replaced by an ever-changing demand. Urban last-mile deliveries became a critical link of the supply chain that suffers from high volatility the most. As a rule, this phase of the chain is considered the most inefficient one because 50% of shipping costs fall on urban deliveries (Lim et al., 2018). Therefore, the last mile delivery became a source where any fluctuations in consumer demands become evident.

There is considerable uncertainty in orders in last-mile delivery. Demand volatility in the chain varies seasonally (because of holiday cycles and promotion activities), daily (as the result of consumers' shopping behavior) and short-term (caused by any unexpected external factors, like social media campaigns or the increase in the number of orders from certain vendors) (Savelsbergh & van Woensel, 2016; Suzuki et al., 2025). Demand volatility could therefore be divided into two groups – predictable cycles (such as the peak during the Christmas time and weekly peaks) and unexpected changeable elements (Pahwa & Jaller, 2023). In spite of the inability of parcel carriers to cope with the high volume of parcels with the existing capacity, operations managers and coordinators are forced to deal with demand volatility in order to ensure the process flow.

Although many scientific articles discuss the long-term consequences of increasing e-commerce on strategy making and financial performance, the operational aspect of last-mile delivery is frequently neglected (Gevaers et al., 2011; Masorgo, 2023). Currently, there is little information about the experience of daily demand volatility in the perspective of managers, coordinators, and drivers, their decisions and options, as well as their consequences. The purpose of the current research is to bridge the gap.

1.1 Problem Statement

The core difficulty in delivering orders faced by the providers of parcel services lies in the

contradiction between physical and human limitations and high volatility of demand. Sortation facilities are limited in capacity, vehicles have restrictions in load capacity, and personnel require a certain level of qualification and working environment (Lim et al., 2018). However, providers should be able to function in both high and low-demand periods without fast-growing possibilities. Each of these periods brings specific problems and costs for companies operating in urban environments. High demand implies exceeding available capacities and consequently leads to traffic congestion, delayed deliveries, and increased working load. At the same time, low demand makes it necessary to keep certain resources idle. These challenges are intensified because of traffic jams and restricted areas of delivery in city centers (Savelsbergh & van Woensel, 2016). It means that understanding how these problems appear in operational terms and what solutions can be offered to overcome them is important both for providers of services and scientists who study them.

1.2 Purpose of the Study

The current research intends to analyze the impact of demand variability on the operations management and decision making of Swedish providers of parcel services. Many academic papers deal with the problem of last-mile delivery. However, their focus often tends to be shifted towards strategic issues (Gevaers et al., 2011; Núñez-Merino et al., 2025). Therefore, the goal of this research is to find out how demand variability affects the operation management and decisions of providers of parcel services in terms of routing, scheduling, and allocation of resources.

1.3 Research Questions

How does demand variability influence the operational management and decision making in parcel service providers? To answer this research question, it is necessary to explore the following interconnected questions: (1) How does variability in parcel volume affect routing, stop density, and resource allocation on each shift? (2) What problems do providers face in the period of peak demand and low demand, and how do these problems differ? (3) What methods do managers use to cope with demand uncertainty, and how do they involve flexible staffing, outsourcing, and re-routing? (4) Do these measures create additional difficulties? Answers will be revealed based on the results of qualitative analysis of semi-structured interviews with operations managers of UPS, Airmee, Bring, and PostNord.

2 Literature Review/ Theoretical framework

2.1 Literature Review Methodology

The literature review was conducted through the use of high-quality academic journals, reports and e-books derived from academic databases such as Scopus, Web of Science, Emerald Insight, Wiley Online Library, Science Direct, Google Scholar and the university library search system Primo. Scopus AI was also used to identify and cite certain scholarly journals.

Researching the topic through various keywords helped to identify scholarly materials which are highly relevant to this study. The primary keywords used included ***"demand volatility supply chain"***, ***"parcel service provider"***, ***"delivery networks"***, "***demand volatility logistics"***, ***"logistics operational management"***, ***"modern logistics strategies"***, ***"delivery tracking software"***, ***"demand uncertainty supply chain"***, ***"capacity planning logistics"***, ***"delivery network flexibility"***, ***"agile supply chain"*** and ***"logistics challenges".*** These keywords were used together with Boolean operators to accurately identify the scholarly materials relevant to this study.

All the scholarly materials used in this study were carefully screened to ensure that they are relevant, widely cited, published under well recognized journals and publications, peer-reviewed scholarly materials were given the priority and more recently published scholarly materials were used to comprehend modern logistics and delivery networks.

Backward and forward citation tracking was conducted to identify scholarly materials that had in-depth knowledge on foundational knowledge on logistics and operations and latest research on modern delivery networks and logistics.

Academic integrity has been maintained through this literature review by accurately citing all the scholarly materials and only university library resources and databases were utilized to conduct this study.

Finally, the selected literature was organized into thematic categories aligned with the research framework, including demand volatility, operational challenges, forecasting, and supply chain flexibility.

2.2 Introduction to the Theoretical Framework

This chapter aims to explain the theories and concepts on demand variability and volatility in parcel service operations, specifically how demand variability creates uncertainty in decision-making, operational systems, logistics and management. This literature review is derived from operations management research focused on seven elements which are decision making under uncertainty, parcel network systems, operational risks, management learning, strategic decision making and demand variability and volatility.

The theoretical framework of this study is designed according to academic recommendations which includes the design of the research conducted through multiple theories, a structured data collection and the empirical findings being carefully interpreted. Also, the theoretical framework is designed to test the empirical findings with multiple theoretical perspectives because in parcel service operations there are multiple interacting factors that could shape the results. Therefore, the theoretical framework of this study does not use a single theory as an analytical tool

In addition, this theoretical framework is built on 2 important concepts which are Bullwhip Effect and Dynamic Capabilities. The Bullwhip Effect (Lee et al., 1997) states that when the demand fluctuates it can affect the supply chain and operations and the Dynamic Capabilities concept (Teece, 2007) states that when there are volatile and changing circumstances, how effectively can the company adapt to these changes and how will they respond? Furthermore, this theoretical framework utilizes logistics research relating to urban last-mile delivery, parcel service operations constraints and management coordination (Savelsbergh & van Woensel, 2016; Lim et al., 2018; Pahwa & Jaller, 2023).

The theoretical framework takes into account many concepts and perspectives that are relevant to this study because it will be helpful to interpret the empirical findings and provide more pragmatic solutions and analysis rather than ambiguous explanations.

2.3 Parcel Service Providers: Operational Networks and Structural Challenges

Hub-and-spoke logistics network is commonly used by parcel service providers to ensure packages are delivered to the consumer safely and on time. A hub-and-spoke logistics network has proven to be efficient because all the packages will be consolidated at a central hub before they are being redistributed to their respective destinations. Such a network can handle large volumes of parcels by reducing the average costs which can generate economies of scale and increase profitability.

However, a drawback of such a network is that there are dependencies between different parts of the network which makes the whole system very vulnerable (Campbell & O'Kelly, 2012). This means that if one part of the network is disrupted, then the whole network will crash (Leskinen, 2021). For example, if there are delays in the transportation of the packages to the central hub then the sorting facilities will be congested, deliveries have to be rescheduled and the staff will be confused and under pressure.

Another disadvantage of this network is that for it to be operational, the company must invest heavily on infrastructure, machinery needed for the sorting of packages, a fleet of vehicles and fully trained staff to handle the operations. Such structural rigidity can cause disruptions to the whole network when the demand is volatile and will be incapable of responding to fluctuations in demand (Cooper et al., 2019).

Additionally, parcel service providers are under pressure to ensure that the packages are delivered to the consumers on time. So, another drawback of this network is that when there are capacity constraints or demand fluctuations, the parcel service providers are unable to deliver the packages according to the schedule which can cause disruptions to the whole network, resulting in customer dissatisfaction and losses (Pahwa & Jaller, 2023).

Also, parcel service providers are affected by external factors such as regulations, traffic, congestion, adverse weather conditions and limited parking availability (Gevaers et al., 2011; Savelsbergh & van Woensel, 2016). So, if such a network cannot handle demand volatility and variability effectively with flexibility, then such external factors will have significant negative impacts on the parcel service providers.

Overall, parcel service providers must comprehend that a hub-and-spoke logistics network during demand volatility will be a significant operational challenge.

2.4 Demand Variability and Volatility in parcel service providers

Due to rapid technological innovations, e-commerce and changing customer behaviors, there is a significant increase in demand variability and volatility in parcel logistics and operations. In such a modern era, parcel service providers must be prepared to handle unplanned volumes of packages and individual customer orders (Savelsbergh & van Woensel, 2016).

When the demand is volatile, parcel service providers can predict certain external factors that cause such volatility. Forecastable external factors include seasonal demands and promotional campaigns. But there are also unpredictable factors such as adverse weather conditions, delays in transportation caused due to traffic and congestion, unexpected disruptions to the supply chain and sudden increases in online orders (Galkin et al.,2025).

To handle such unpredictable factors, the Bullwhip Effect provides useful insights on how to overcome demand variability. The Bullwhip Effect explains that when the consumer demand increases the parcel volume will increase. So, there must be buffering mechanisms in place so that the sorting facilities are prepared to handle such an increased volume of packages and the company must make quick decisions and respond quickly to the incoming orders and increased demand (Lee et al., 1997).

But when there is an increase in demand which results in an increase in the number of deliveries, there is a short time period available to adjust the parcel delivery system to these changes (Suzuki et al., 2025). During this short period of time, the whole parcel delivery system can overload and crash and the management at the operational level will be negatively affected.

Furthermore, parcel delivery providers can use historical data to prepare forecasts and assessments to effectively handle demand volatility and its challenges. However, such forecasts must be equipped to predict unexpected deviations. Therefore, there must be a balance between relying on forecasts but also being prepared and flexible to deal with real time sudden changes (Pahwa & Jaller, 2023).

Overall, unpredictable factors and deviations pose a challenge for parcel service providers. They must be prepared with a coherent plan to deal with both expected demand and unexpected changes. This issue is central to comprehending the operational and managerial challenges discussed in the following sections.

2.5 Operational Challenges Arising from Demand Volatility

Demand volatility negatively affects parcel service providers and many parts of the parcel service operations. The significant component of parcel service operations are the sorting facilities where all the packages arrive before they are being delivered to the consumers. In the event when there is an increase in demand, the sorting facilities will be heavily under pressure as they are required to process large volumes of packages within a limited time and when the number packages exceed the limit they can handle, the whole process will lead to bottlenecks, congestion and delays (Pahwa & Jaller, 2023).

If the sorting facility cannot handle the volume of packages, the parcel service providers will have to delay the processing or redistribute the packages to an available sorting facility which can increase costs and affect service levels. On the contrary, if there are less packages with low demand then there will be an underutilization of resources which can lead to high costs per unit and diseconomies of scale (Cooper et al., 2019).

Furthermore, when the demand increases there will be more pressure on the staff which can cause exhaustion and mental strain. Such a circumstance will lead to trade unions to sue the parcel service providers for breaching labour laws or parcel service providers have to pay for overtime work which can increase costs. Therefore, flexible working arrangements with proper coordination and team work must be introduced to mitigate the additional strain on the staff (Auad et al., 2026).

Additionally, drivers and transportation will be another problem during demand volatility. When there is a large volume of packages that need to be delivered, there must be additional vehicles and drivers available if not deliveries will be delayed and customer retention will fall. On the other hand, if there is low demand then the excess capacity of vehicles and drivers will lead to inefficiency. Therefore, parcel service providers must arrange flexible schedules and temporary/renewable contracts with the drivers to ensure that the delivery system runs efficiently (Leskinen, 2021).

Also, during such a demand volatile situation if deliveries have to be made in urban environments the drivers will have to deal with traffic, congestion, regulatory constraints and limited parking availability which can delay the delivery of the packages on time to the consumer (Gevaers et al., 2011). So, in such a situation the parcel service providers must guide the drivers with flexible route planning and coordination.

In Demand volatility helps the parcel service providers to identify the flaws in the delivery and logistics networks and they need to focus more on making the system more flexible through the use of adaptive strategies that can balance capacity utilization with flexible decision making.

2.6 Managerial Responses and Adjustment Strategies

Through extensive research it was discovered that managerial responses and adjustment strategies can be deployed to reduce the impact of demand volatility and these include a flexible workforce, terminal proximity to the clients, the use of technological innovations and horizontal collaboration. This study also identifies the limitations of these managerial responses and adjustment strategies.

A flexible workforce can help to deal with demand volatility and this can be achieved by offering the employees different types of employment. This means that parcel service providers must hire a mixture of fixed employed workers, temporary/part-time employed workers and they must also hire an employment agency who has employees on standby for recruitment (Auad et al., 2026). Using an employment agency is highly beneficial when there is unpredictable demand volatility (Zhu et al., 2025). But such employees aren't familiar with the sorting facilities, the parcel delivery system and route familiarity. So, the parcel service providers must provide some induction and training to these external workers which can increase incompetency and costs. Also, when there is less demand and less volume of packages then they have to pay for every fixed/permanent employed worker even if they are under-utilized (Auad et al., 2026). Therefore, the parcel service providers must invest in developing a system that will be capable of undertaking quick mobilization and deployment of flexible workers (Pahwa & Jaller, 2023).

Technological advancements can also help with demand volatility such as investing in a software that quickly updates, informs and makes real-time adjustments on the volume of

packages, operational conditions and delivery routes (Liu & Wang, 2026). But even such a software has constraints such as traffic and congestion will be hard to assess, adverse weather conditions are hard to predict accurately and if the software crashes the whole delivery system will lead to chaos and losses.

Additionally, many parcel service providers rely on machine learning algorithms rather than statistical models because algorithms are more accurate, forecast massive data related to the volume of packages and they provide quick responses in terms of planning and scheduling of labour and vehicle routes (World Journal of Advanced Engineering Technology and Sciences, 2025; Aylak, 2025).

Another, crucial technological innovation that can help with demand volatility is advanced dynamic vehicle routing techniques. This tool can accurately adjust and update the delivery schedule and the delivery routes based on new volume data received (Liu & Wang, 2026). This is very beneficial to improve the coordination with the drivers and ensure that the packages are delivered to the correct location on time. But such tools cannot replace the experience and the judgement of human managers as managers can account for unexpected events and deviations (Núñez-Merino et al., 2025; Suzuki et al., 2025). Therefore, there must be a coordinated effort and an examination between the managers and the decisions made by the advanced dynamic vehicle routing techniques and this accounts to be one of the most important contributions to this research.

Furthermore, if there is an increase in demand and an increase in the number of packages to be delivered, the parcel service providers can outsource transportation or collaborate with another parcel service provider and this is called horizontal collaboration (Argyropoulou et al., 2023). The 3 main advantages of using third-party carriers are that they are flexible, can increase service levels and that they no need to invest money on buying vehicles, hiring new drivers and building the necessary infrastructure (Zhu et al., 2025). But there will be high dependence on external carriers, high fees and poor coordination with the delivery system.

The Dynamic Capabilities (Teece, 2007) concept helps to comprehend these strategies more accurately and it emphasizes that demand volatility cannot be eliminated as it gives rise to second inefficiencies such as increased operational costs and problems with coordination.

2.7 Decision-Making Under Uncertainty

To ensure the delivery system and the operations are performing without any problems, the managers must make quick decisions even without referring to forecasts and accurate data. Therefore, this is against traditional models of rational decision-making.

Decision-making involves the use of analytical tools and wise judgement. But most managers rely on experience and intuition to make decisions because these analytical tools and forecasting systems don't always reflect the complexity of real-world problems (Núñez-Merino et al., 2025). Another reason why managers rely on experience and intuition is because the environment of parcel delivery and logistics is highly dynamic with continuous changing situations. Managers must always be informed and updated on the delivery operations so that they can comprehend the situation and adjust their decisions accordingly.

Effective communication between the manager and the drivers, staff, organizers, planners and coordinators is also crucial because information must be shared quickly and accurately so that the manager can make well informed decisions.

Overall, in a situation when the demand is volatile and there is uncertainty, the manager must not rely solely on experience and intuition but must also study the forecasts and data available.

2.8 Planning vs Execution: The Operational Gap

A significant problem when managing a delivery system and its operations is that there is a gap between planning and execution because in the planning stage the manager relies on data, forecasts and structured models but at the execution stage actual operations are subject to real-time changes and disruptions (Simchi-Levi et al., 2008).

The environment of delivery systems and logistics are subject to very high levels of volatility and variability because there are many unpredictable factors that can disrupt at any time the entire delivery system and operations and this is an inherent characteristic of delivery systems and logistics (Christopher, 2016). Therefore, in such an environment, forecasts and the data are not always reliable. This creates a huge wedge between the planning stage and execution.

Furthermore, another reason why there is a gap between planning and execution is due to information asymmetry (Michalski et al., 2018). This means that planners have access to forecasts but they don't have much information on detailed real-time data on operational conditions because there are continuous updates and adjustments that have been made to the system and its operations.

This gap between planning and execution demonstrates that in logistics systems a more flexible and adaptable approach must be adopted because traditional planning methods aren't very effective.

2.9 Organizational Learning and Adaptation

The pragmatic approach that delivery service providers can adopt to deal with demand volatility is through organizational learning. This means that when the delivery service providers have long experiences in managing delivery systems and its operations, it is knowledge based on experiences that delivery service providers have accumulated over the years which will be beneficial to draw upon personal experiences and reflection to deal with the issue of demand volatility.

Delivery service providers can accumulate organizational learning through formal and informal processes. Formal processes include data analysis and performance reviews whereas information processes include accumulating experiences, knowledge and strategies and then sharing them among the employees. In the long-term, organizational learning can help to make well experienced decisions and delivery service providers will be well equipped in dealing with demand volatility.

Therefore, organizational learning can aid delivery service providers with a unique set of skills based on real-time experiences and practices which will improve the resilience of the delivery system and its operations (Teece, 2007).

3 Methodology

3.1 Research Design

In terms of the methodology, this research relies on a qualitative case study design intended to address the issue of how fluctuations in the level of demand impact operations

management and decision-making in Swedish parcel delivery companies. Qualitative research is an appropriate option in this case as the study is focused on understanding how the managers, coordinators, and drivers perceive changes in the parcel flow and how decision-making happens (Braun & Clarke, 2006). Quantitative techniques may be employed to reveal the magnitude of volume variations; however, in order to understand human-based aspects of operations under demand instability, qualitative research is a preferable choice.

Considering that the focus of this research is on demand volatility and its influence on operations, the selection of multiple case organizations appears reasonable because the researcher aims to discover if the discovered phenomena are organization-specific or typical for the whole industry. For this reason, UPS, Airmee, Bring, PostNord, and Bonway Logistics are selected. The organizations function in similar conditions when it comes to geographic setting and regulatory requirements although they differ in many aspects such as services provided and types of customers served.

As far as the research design is concerned, it is based on the conceptual framework discussed in Chapter 2 where the interrelation between demand volatility, operational problems, managerial actions, and decision-making is explored.

3.2 Data Collection

3.2.1 Primary Data

Primary data collection was achieved by means of conducting eight semi-structured interviews with the participants actively involved in the terminal and delivery operations within five parcel delivery firms. For this research semi-structured interviews have been deemed most appropriate, as they enable probing for further depth of the participants' experiences while still maintaining some degree of structure in the interview process, which would facilitate both within-case and across-case analyses. Moreover, semi-structured interviews allow the participant to bring up new themes of discussion beyond the scope of pre-set questions.

The selection of interviewees is based on representation of a variety of roles involved in the processes of parcel sorting and preparation in the terminal operations and in delivering packages, so that the influence of volatility will be traced throughout all stages of the

processes. Participants include:

Table 1: Overview of Interview Participants

Role	**Organization**	**Key Responsibilities**
Driver / Route Operator	UPS	Independent sorting, loading, and execution of B2B delivery routes
Site Manager	Airmee, Jönköping terminal	Operational planning and coordination of last-mile deliveries
Operations Traffic Coordinator	Bring, Jönköping terminal	Management of intermodal flows, truck booking, and night shift communication between terminals
Production Manager	Bring, Jönköping terminal	Sorter-side production, staff management, and peak planning (up to 70+ employees)
Terminal Worker	Bring, Jönköping terminal	Sorter and cross-docking production, loading/unloading, pallet scanning and handling
Team Leader (Acting)	Airmee, Jönköping terminal	Night shift operations, inbound/outbound coordination, sorting supervision, and acting leadership
Area Manager	PostNord, Stockholm	Management of B2B pallet and parcel deliveries via six local managers
Distributor Driver	Bonway Logistics, Jönköping	Parcel and newspaper delivery during night shifts across districts

All interviews have been conducted in English language and were recorded with participants' permission. Later transcripts have been prepared. Prior to the interview, participants were informed about the purposes of the research and of their right to withdraw at any stage. Moreover, all data is processed anonymously.

3.2.2 Secondary Data

Secondary data include industry reports, operational performance documents and academic literature on last-mile logistics and demand management. These documents are utilized for the purposes of contextual analysis of the interview data, creation of the theoretical background in Chapter 2 and triangulation, when convergence is observed between the data collected and industry practices.

3.3 Sampling Strategy

A **purposive sampling** method was used to select participants based on their involvement in operational decision-making within parcel service activities. Purposive sampling ensures that the data collected is relevant to the research objectives and reflects practical experience with demand volatility.

The sample included **8 participants**, distributed across managerial and operational roles. This size is sufficient to achieve data saturation while maintaining depth of insight. The inclusion criteria are:

- Direct involvement in daily or strategic decision-making related to delivery operations.
- Minimum one year of experience in parcel delivery or logistics management.
- Willingness to discuss operational challenges and decision-making practices.

Participants were contacted through company networks and professional logistics associations. Interviews will be scheduled at times convenient for participants, either in person or via online platforms such as Microsoft Teams and Google meets.

3.4 Interview Guide

The interview guide contains a set of topics structured according to the six themes developed on the basis of the literature review and research questions. Every topic is related to one or several sub-questions and was used in all eight interviews in the same way, but the follow-up probes were adjusted according to the respondents' position. The themes are: demand volatility experience, operational challenges, Managerial responses to and adjustment to volatility, decision-making under uncertainty, the difference between the planning and the

execution levels, and organizational learning and future adaptation. The complete guide is included in Appendix A.

3.5 Data Analysis Approach

Interview data were analyzed using the thematic analysis method suggested by Braun & Clarke (2006) and encompassing six phases: familiarization with the transcripts, code generation applied to key phrases and passages, clustering codes into broader themes corresponding to the framework, review and revision of the emerging themes, and presentation and illustration of findings with representative quotes. As a deductive method, thematic analysis uses the pre-existing themes developed from the conceptual framework and formulated in the interview guide as an overarching framework guiding the interpretation process. It was chosen because the research questions emerged out of a conceptual framework, and thus, it would be appropriate to explore theoretically defined concepts such as demand volatility, operational flexibility, planning-execution gaps, and decision-making under uncertainty through real-world experiences.

Thematic analysis was applied based on coding the interviews deductively according to the six themes of the research framework. Triangulation was conducted on two levels: across roles and across organizations. Findings related to each theme were cross-checked according to the drivers', coordinators', site managers', production managers', and area managers' perspectives and assessed for their consistency across the different vantage points and for their dependence on the respondent's position. In addition, patterns observed across several organizations-such as the significance of experiential judgment in decision-making or asymmetry of challenges at peak versus trough periods-were considered more substantiated.

3.6 Research Quality and Limitations

In order to provide for high-quality research, it relies on four criteria: credibility, transferability, dependability, and confirmability. First, credibility was achieved through triangulation and direct quoting of participants. Second, transferability was attained by presenting the results in the context of the Swedish parcel delivery sector to allow for evaluation of their relevance for other sectors or countries. Third, the interviews were conducted and analyzed using an explicit protocol, which guarantees dependability. Fourth, credibility was achieved through data-based findings and clear differentiation of observations

from interpretations.

The major limitation of the research relates to the limited sample size. Although eight interviews with five companies can provide enough data for qualitative insights and for thematic analysis, they cannot be used for making any statistical claims regarding the phenomenon. Thus, the findings should be regarded as analytically generalizable for other similar operating environments rather than statistically generalizable to all cases. In addition, focusing on only Sweden means that some aspects specific to the Swedish labor market, sustainability policies, public holidays, and urban delivery system may not be present in other national contexts. Moreover, all five companies have been operating for many years and are relatively large providers; thus, small or new operators might experience volatility differently.

3.7 Ethical Considerations

All participants have been thoroughly briefed about the purpose and goals of the research prior to the interview. Participation was entirely voluntary, and it was explicitly mentioned to the interviewees that they can leave the interview whenever they want and do not need to provide an explanation. Anonymization of participants was ensured: names and company identification markers were not included in the interview transcripts or the appendices. Data security is assured, and interviews were used for academic purposes in accordance with GDPR legislation and ethical standards at Jönköping University.

4 Findings

4.1 Overview of Participating Organizations and Respondents

The empirical data reported in this chapter has been collected through eight semi-structured interviews with operational personnel from five parcel delivery organizations in Sweden: UPS, Airmee, Bring, PostNord, and Bonway Logistics. These respondents were involved in roles ranging from front-line route operators, terminal managers, site managers, and area managers, thus ensuring the representation of multiple operational hierarchies of the parcel delivery.

The respondent at UPS was a driver and route operator responsible for independent sorting, loading, and executing a dedicated B2B route cluster. At Airmee, the interview included both

the Site Manager of the Jönköping terminal, responsible for operational planning and last-mile coordination, and the Night Shift Team Leader, responsible for managing inbound and outbound trucks as well as the night shift production. At Bring, the interview included both the Production Manager in charge of sorter-side production and workforce management, and the Operations Coordinator managing intermodal traffic flows, and the Terminal Worker conducting loading, unloading, scanning operations, and internal process improvement projects. At PostNord, the respondent was an Area Manager responsible for the Stockholm logistics region. At Bonway Logistics, the respondent was a night-shift distributor with about nine years of experience of delivering parcels and newspapers across district routes in and around Jönköping.

Bonway Logistics provides a different organizational structure among the five organizations considered in this study, in the sense that the company is not employing a hub network with sorting and terminal management infrastructure. Rather, Bonway uses the night-shift model, whereby drivers operate independently in a given geographic district without being responsible for sorting at the terminal. For two companies (Airmee and Bring), two respondents have been selected for interviewing in order to capture both the management- and operational-level perceptions of the same process. Overall, the eight respondents represent the role categories mentioned in the sampling frame: terminal production management, delivery performance coordination, route planning and traffic management, last-mile delivery execution, frontline terminal operations, dispatcher coordination, and autonomous night-shift distribution. The empirical results are structured according to the six thematic areas outlined in the interview guide and related to the research questions.

4.2 Experience of Demand Volatility

4.2.1 Seasonal Patterns

Demand volatility has been identified as a key feature of the organizational context among all eight respondents in this study, in terms of two major temporal dimensions: seasonality and intra-week demand. In terms of seasonality, the period between September and December was reported by all organizations as the period of the highest demand increase. According to the UPS driver, demand increases by "about 300 percent" in the fourth quarter, especially "two weeks before Christmas," which requires ten-hour shifts to cope with demand. The

Bring Production Manager provided a clear picture of the magnitude of seasonal demand variation: "When we started, we had a volume, maybe on an average ten, twelve thousand that's like the best volume that we had per day. And then now we get maybe seventy, sometimes eighty thousand even in the low season, and during high season it goes up to sometimes one hundred twelve, one hundred fifteen, even one hundred twenty thousand sometimes." The Bring Terminal Worker confirmed this observation, citing "November because of the Black Friday bills, Christmas presents and all that" as definitively "the peak where you have the most volumes," and adding Easter as a second peak. The Airmee Night Shift Team Leader highlighted the operational consequences of the holiday period: "If you have Christmas and if you have Easter, any other vacation coming up, we can always have an unexpected volume coming into the warehouse, so in such cases we need to always be prepared."

The reverse seasonal demand pattern was equally clear. According to the UPS respondent, demand could decrease "by a factor of two or three" after the midsummer period, as major manufacturing clients suspend their activity during the industrial vacation period. The Bring Terminal Worker attributed the summer period of low demand to non-commercial rather than commercial reasons: "People are going for vacations, most companies are understaffed, so it's really lax demand for products, and people are actually more outside." The Airmee Site Manager reported that winters cause overstaffing in the organization, since "in the winter I would say out of the seven days, at least four days we are overstaffed" due to trucks that are delayed by snow and blocked roads. The PostNord Area Manager observed an interesting phenomenon with regard to macroeconomic seasonality: "During hard economic times, instead of buying a television for a thousand euros, you bought a toothbrush and a speaker for your phone for maybe five hundred euros so instead of one parcel there was two."

4.2.2 Intra-Weekly Patterns

Each firm possessed a unique weekly demand curve, influenced by their particular approach to services and consumer group characteristics. According to the UPS driver, "on account of corporate organizations closing down at weekends, the build-up of B2B freight is bound to lead to a great deal of operational pressure on Mondays and Tuesdays." Under normal circumstances, the average number of parcels handled per day was estimated to be approximately sixty units, increasing to eighty on Mondays and dropping to forty to

forty-five on Fridays. The PostNord Area Manager described a reverse consumer-driven pattern: "We have a lot of incoming goods on Monday, Monday afternoon, Monday evening. We sort it out in the terminals on Monday evening. We ship it through the country, and on Tuesday we have the highest peak of deliveries. Absolute highest peak." The Bring Terminal Worker's description corresponded to the consumer-driven trend: "At the beginning of a week it normally feels a lot - on Mondays and Tuesdays. I think that's because over the weekend so many orders are going in and being put in. Then as the week progresses it actually gets calmer and calmer." The Airmee Site Manager pointed out Tuesday, Wednesday, and Thursday as a high-demand period, whereas the Airmee Night Shift Team Leader described the situation as follows: "Throughout the week, Wednesday we have recognized as the day where we get the highest," although he qualified this statement by noting that "it keeps changing" due to client delivery schedules. The Bonway distributor described Tuesday through Friday as a high-volume period for parcels, whereas Monday was considered the time for delivering newspapers only and took up to four hours.

4.2.3 Drivers of Fluctuation

Seasonal promotional campaigns and holiday peaks were the major demand fluctuation factors. However, the respondents also revealed a wide range of other causes, depending on their operational context. The Airmee Night Shift Team Leader pointed out the complexity of demand factors: "There are a lot of factors that come into play, like the seasonal peaks, and then there are discounts with the clients, and then the customers' expectations and regional demands." He stressed that particular types of clients caused uncertainty as far as demand forecast was concerned: "We always have these clients with unexpected volume from Arcanus, Zooplus, and Apotek, because people in Sweden have a lot of pets" and visit pharmacies often, hence the difficulty in forecasting. The respondents pointed to Amazon as the major client factor: "They have sales going on some days where sometimes unexpectedly we see a lot of high volume coming in." The Bring Terminal Worker emphasized consumer-related factors as the reasons for short-term weekly fluctuation: "Over the weekend people are free and have downtime, so they're on their phones ordering stuff - that's where you get that spike at the start of the week." The Bring Operations Coordinator mentioned winter-related disruptions such as snowstorms, road closures, and ferry cancellations as a considerable secondary cause of line-haul volatility. The Airmee Site Manager identified transit problems caused by the retailer as a more significant factor than consumer demand.

The former made up for "around twenty to twenty-five percent of what we miss," while the latter accounted for "five to ten percent of missed deliveries." The Bonway distributor specifically ruled out traffic as a factor ("In the night there is no traffic, so you just work") and mentioned the winter season with adverse weather conditions (snow, ice) and the risk of being stuck on the road as the main factors causing delivery disruption.

4.3 Operational Challenges During High-Demand Periods

High-demand periods generated unique capacity constraints, depending on the respondent's operational level. At PostNord, the sorting capacity was the major constraint, according to the Area Manager: "We had at least two to three terminals at their absolute maximum capacity" in the month before Christmas. About 80 percent of parcels were delivered between 16:00 and 00:00, meaning that the company had a four-to-five-hour window for handling parcels. The physical constraints related to sorting facilities (fixed floor space, fixed machine capacity, and difficulty in adding extra workers due to tight space) meant that increasing labor did not result in increased efficiency. The standard response was to ask major clients to start deliveries earlier: "We tend to inform our clients that we have to come earlier-whether they like it or not, or they won't be able or we won't be able to meet the quality measures they desire."

For the Bring Terminal Worker, the challenge associated with high volume at the terminal level could be described as departure planning: "I think the biggest challenge comes with the line haul departures and planning. It's very difficult when you have a lot of goods, packages, pallets piling up, but then you don't really have any departures, then you're going to have congestion, and probably once from there, when with that congestion, then you get slower, then in case a line haul departure arrives, then you have stress to kind of load it as soon as possible." Such a description of how congestion results in slower sorting, which leads to rapid loading, illustrates the sequence of consequences of volume surges on the terminal floor level. At UPS, capacity constraints could be seen at the route level: when driver loads exceeded route capacity (90 parcels against baseline sixty), management distributed inventory among other drivers. In case of saturation of the entire network, the standard procedure was to deliver as much as possible and take back the rest to the terminal. As for the most complicated category of operational challenges, the interviewee emphasized parcels with delivery window requirements: "There is a noticeable psychological strain on personnel

during high-demand periods. Operationally, the most complex challenge involves parcels with specified delivery windows-narrow temporal slots that must be met regardless of total volume." The PostNord Area Manager gave a clear example of service level ceiling: "I would say that ninety-five, ninety-six percent should be good enough, because if you go above ninety-seven, ninety-eight percent it becomes so expensive that you won't be able to do business." According to the Bring Production Manager, the peak periods put the greatest strain on human resources: "The biggest challenge is about the people-the resources that we have, because during peak the volume comes crazy from all the corners." He gave an example of how recently the network failure at the Gothenburg terminal at 15:30 forced him to change the production plan for the next shift starting at 06:30: "I had to, you know, like, really act on that second, on that very second." The Airmee Night Shift Team Leader explained the strategy his company used to respond to staff pressure: "When we see this amount of staff cannot handle the volume, then we try to book in more employees during the day, even if the shift has already started, we try to add in more employees, because we don't want to put on more stress, because it's a type of work where people have to move things physically." The Airmee Site Manager confirmed the sustainability-oriented carrier capacity constraint by emphasizing that "At this point in Sweden, there aren't many companies that utilize what we need them to use. So the ones we already use are already pretty maxed out on days when these kinds of disruptions occur." He also mentioned the company's practice of stopping accepting new orders when the load volume exceeded forecasts to a certain extent: "Let's say if it is thirty percent above forecast, that's something we can do. If it is something like seventy percent above forecast, we put a hard stop on forty percent. The remaining thirty percent is another day's problem."

In general, the most common immediate operational response to volume surges was route density adjustment. At UPS, surplus parcels were redistributed among adjacent drivers in the morning load-in. The PostNord Area Manager emphasized the importance of continually filling the space: "Efficiency is everything. You have to fill the space." The Bring Operations Coordinator gave the most detailed description of how route splitting and reassignment decisions were made, involving real-time assessment of driver working-time restrictions, customer closing time, distance to destinations, and trailer compatibility, resolved through dialogue with terminal employees or drivers within minutes.

4.4 Challenges Arising during Low Demand Times

During the low demand periods, the primary problems stemmed from the high cost associated with maintaining excess capacity. The Airmee Site Manager pointed out that switching from high demand to low demand requires considerable reductions in staff: "The issue here is that there is overstaffing, especially in winter period – on at least four out of seven days." The Bring Terminal Worker highlighted the challenge of maintaining sufficient staffing during low demand periods: "You have some people on contracts, or the full timers – these people have to come and fulfill their contracts. So if you have really low volumes, then you are faced with that challenge – you have a lot of people and you have to have them at work, but then you have nothing for them to do. So still they are paying out wages for no work being done." He viewed this challenge as two-way: "It's in both scenarios" – the high demand creates overload, while the low demand creates idle cost.

A more optimistic outlook was provided by the Airmee Night Shift Team Leader, who argued that "when you have steady volume, challenges are not always something that we face day-to-day" and that "during low volatility the prognosis always matches with the actual demand." However, he agreed that the distinction between low and high demand is quite unpredictable: "We actually thought that once Christmas was over we were going to expect a low volume, but all of a sudden, unexpectedly, people started buying things." PostNord Area Manager described a differentiated solution to the low demand problem, with permanent B2B drivers forming a stable core, while part-time student labor flexibly scales the B2C evening operations; moreover, "in Stockholm and the major cities we also have Saturday deliveries" – while the constraint during peak season was switched from staffing to vehicles. The UPS respondent and Bonway distributor did not mention low demand periods as a relevant operational challenge.

4.5 Managerial Responses and Adjustment Strategies

Staffing flexibility emerged as the universal adjustment strategy among the organizations with a formalized management structure. The Bring Production Manager kept an active labor pool specifically aimed at adapting to demand fluctuations, which was developed over four to five years: "On a daily basis we don't rely on subcontractors. We have our own demand pool – whenever we know that the volume will go down, we reduce the people. Whenever we know the volume will go up, we increase it." He stressed the importance of having a reliable

inner circle: "We always have some backup – whoever is living nearby. We always have some of them that always can come." The Airmee Site Manager relied on the most flexible staffing approach possible, pointing out that "We have temporary staffing throughout our operations to meet the volatility not just in peak but also day-to-day activities – so it's more or less everything is temporarily staffed." The Airmee Night Shift Team Leader discussed his organization's two-tier approach: "We have employees that work for Airmee, and then we have demanding – just like in other warehouses – from Temply and Flexio, and those workers get updates to an app called Temply where the shifts are published, and people can pick up the shifts."

The Bring Terminal Worker provided details regarding Bring's shift from relying on external agencies to developing its own labor pool: "Before, they used to have other providers, for example Uniflex, but then they developed their own pool of staff – so these are within Bring themselves, not any external company providing extra staff." It appears that it was a purposeful strategic decision.

Route and traffic adaptation included the immediate adjustment of delivery plans and line-haul schedules. The Airmee Night Shift Team Leader mentioned rerouting even outside the warehouse premises: "Right now we have constructions going in Rosenlund, one in Grammar, so at these times when the courier drivers are delivering their products, when they see these constructions happening, they have to reroute. So then we use the reroute optimization scenario." Bonway distributor described an autonomous variant of the same strategy: although the delivery application suggests a certain stop sequence, "it doesn't really depend on you – you can start from the ending and finish at the start. Doesn't matter. You make your own decisions because you are the one who is in the area. You know the shortcut."

External carrier management was depicted as highly important, but also subject to structural constraints. The Bring Operations Coordinator indicated that carrier contracts imply booking commitment, with the typical policy consisting of monitoring destination-level volume trends over several days before contracting additional carriers. In the case of Airmee, facing the pressure of reducing the carrier list due to sustainability considerations, the diversification towards European carriers was mentioned as the organizational response. Both Bring respondents emphasized that the organization adopted the culture of volume absorptiveness as a strategy: "We never say no to any volume that is coming to us," according to the

Production Manager.

4.6 Decision-Making Under Uncertainty

Across all eight interviews, one key commonality appeared to be that experiential judgment works hand-in-hand with formal data in decision-making. For instance, the PostNord Area Manager, with his logistics experience spanning over three decades, considered gut feeling to be the main tool for analyzing volume trends: "My gut - gut feeling. I have thirty years working with lorries, deliveries, and things like that." However, he still noted an increase in the importance of data analysis: "We have an extreme amount of data - when does the parcel come in, when do we sort them, when do we deliver - so we have millions and millions of data points that we have to harvest." Similarly, the Bring Production Manager noted a combination of data and feelings: "We combine this feeling and real time data - for example, what we had last week, how the trend is going from last few weeks, and then we roll it out."

The Airmee Night Shift Team Leader made a clear distinction between algorithmic and human judgments in decision-making: "AI, yeah, for sure we use it, but human judgment has been majorly involved in the decision-making process." He offered an illustration: "If it's all ten pallets of wine, the AI tool is not going to think that these two people can't do ten pallets of wine because it's too heavy - it's just reading the data. But then when human perspective plays in, we think, no, we need three or four people for this." The Airmee Site Manager also noted that forecasting is, in essence, an analysis of the pattern: "We always try to read the pattern - if it's the season coming up we are always going to be prepared for unexpected volume." Moreover, he mentioned that forecasting gives "ballpark" figures, which "we can definitely rely on and use to manage the staff and the production - it's not exact, but it always gives you a ballpark figure."

The Bring Operations Coordinator viewed decisions as a "fingertip feeling" – tacit knowledge gained through extended experience of the same route, the same carriers, the same volume patterns. However, he noted that sometimes decisions have to be made even without real-time destination-level data, which makes analytical decision-making structurally impossible. The Bonway distributor offered a vivid example of fully autonomous driver-level decision-making: "You take your own decisions because you are the one who is in the area. You know the area, you know the shortcut - instead of going there and coming back again to the same place, you can decide to do it at the same time together."

The pace of decision-making was also perceived as quite pressing. For example, the Bring Production Manager described cases requiring mobilizing resources within one-two hours after being informed about the disruption, with the whole shift's production plan having to be changed accordingly. The Airmee Night Shift Team Leader described an equally urgent pace: "If you have uncertainty volume coming in, the day shift team leader comes early to scrutinize the entire warehouse and informs the manager that we will need more stuff for the production, and then we try to bring them as quickly as possible, because the production day shift divides into some certain areas." The Bring Terminal Worker pointed to the importance of information timing: "The ideal situation will be to act as soon as possible. If this information comes in early, maybe at around three o'clock, it gives management a chance to act on that and put in extra staff. But if this happens during production, like maybe at eight, that is really hard to deal with - so it depends on the timing of when that information comes in."

4.7 Planning-Level Versus Execution-Level Perspectives

Forecasting tools and their reliability differed across the organizations. For example, the Airmee Site Manager reported a 95 percent forecast reliability for daily volume, estimated from historical data, year-over-year growth and retailer communication, created by the solutions team and checked weekly. The Bring Production Manager described the machine-learning prognosis model to be usually quite accurate, with up to three thousand parcels of deviation being acceptable and deviations of ten thousand or more being disruptive. The Airmee Night Shift Team Leader, however, noted the limitations of forecasting: "The prognosis is not always accurate as it's been shown, but we always try our very best to somehow match with it - but there are significant margins." In his view, the uncertainty of forecasts is inevitable in this domain: "The actual volume is always not the same as how it's been shown or said."

The Bring Terminal Worker offered a first-hand insight into forecast reliability: "Some days it could be almost close, but we have days where it's way off - so it keeps varying. That's why the staffing level is always changing. You can't really fully depend on it alone - you depend on that, but then you also have to look at what's actually in the terminal, what you have, then you make a decision based on that." PostNord reported a long-term pallet forecast reliability with deviations of less than half a percent, and daily parcel volume forecast reliability of 2-3

percent per week, along with a project aiming at developing daily prediction capability. At UPS, no forecasting tool was mentioned; however, Monday was considered to be the most unpredictable day.

A notable planning-execution gap also became apparent across all interviews. The Airmee Night Shift Team Leader considered the main gap to be a lack of understanding, which employees should be aware of when expecting uncertainties in execution as opposed to planning on paper: "The biggest gap is the understanding - employees should always fit in with the ethical and cultural perspective of the organization, and always expect some uncertainty when it comes to executing, because planning is something you do on paper, but execution is something where you go to the field and put in the work." The Bring Terminal Worker viewed the exclusion of front-line staff from planning as the main structural gap: "I strongly believe that terminal operators are the process owners - they understand what is actually going on, they know what it takes. But unfortunately, in the planning, they are not really heard a lot. Their advice is not really taken on a daily basis. So that's a huge gap." This points to the information asymmetry occurring not only between retailer and carrier but also inside the organization itself.

The Bring Operations Coordinator considered human factors and mechanical factors to be the two main sources of execution deviations, such as customers loading more pallets than booked, trucks arriving late, underperformance of electric vehicles in cold weather, and ferry cancellations. Larger disruptions resulted in chain reactions impacting a whole week of circular tour vehicle operations. The PostNord Area Manager viewed the planning-execution gap as a structural phenomenon: given daily volumes of close to one million parcels, 95-96 percent service level is economically rational, and "if you go above ninety-seven, ninety-eight percent it becomes so expensive that you won't be able to do business." The Airmee Night Shift Team Leader described the immediate feedback loop in case of problems: "The moment it happens, we inform them, because as a company we take the counter-patent responsibility of delivering the packages and pallets to the customers on time - so the moment something unexpected happens, I try to inform the site manager as soon as possible."

4.8 Communication Structure and Feedback Systems

There was a wide variation in the extent of formal structures used in communication within organizations. PostNord had the most extensive communication system in terms of formal

channels. The Company Intranet was used as the channel for upper-level communication, email and MS Teams for middle-level communication, and recently adopted the MS Teams channel on drivers' production devices. As the Area Manager pointed out, communication often stopped at middle-level managers, with drivers receiving operational orders rather than organizational information. Monthly workplace meetings were supplemented by video recordings available to night-shift workers.

In Bring, the multi-channel communication included the use of Teams channels for written communication, direct phone calls for urgent communication, walkie-talkies for communication between the terminal floor and communication via daily morning and evening meetings. According to the Bring Terminal Worker, the Nexus warehouse management system served as the central communication hub: "In this system, if you are in Jönköping for example, you can have a look at what's happening in Edinburgh – this creates a seamless communication between all these different terminals. It's kind of live data – if in Gothenburg they log in a car, in Jönköping you can access the system and see what they're loading." According to him, the key source of information exchange with the terminals was the line-haul department: "The closest point of contact will be the line haul department – they are the ones in more contact with the technical operations. They have different traffic operators who talk to different customers and try to understand what's going on, so that's how they relay the information."

In Airmee, respondents spoke about a hierarchical communication structure based on open-channel risk broadcasting. In turn, the Night Shift Team Leader said that there was a less formal, but not less intense communication process: "We use different channels – some for ordering extra trucks, some for letting them know if there's any issues, some for getting updates on trucks, and some with a specific routine where team leaders follow a certain form once done with the work. Between colleagues – the night shift team leader and day shift team leader – and with the site manager – we use normal communication methods such as WhatsApp, calls and text messages."

UPS was characterized by the mobile data terminal as the main tool of in-shift communication. Respondents called the terminal functionally obsolete and dated back to the mid-2010s, with navigation problems requiring drivers to use their personal mobile phones. Its modernization was identified as "the single improvement that would most significantly

enhance operational efficiency". In addition, respondents indicated the lack of formal feedback mechanism allowing drivers to add route information to the planning system. In Bonway, the only channel of communication was the delivery application and the only phone for emergencies. Respondents described the absence of team communication, formal feedback from drivers to planners, and structured performance reviews – drivers kept all the route information.

4.9 Organizational Learning and Future Adaptation

Respondents provided examples of various mechanisms of learning from past volatility events. The Bring Production Manager presented the most systematic approach to organizational learning: documentation of decisions, staffing, and results of each peak period using the Lean system, which allows for year-by-year improvements; keeping peak workers in order to keep the experience of past peaks was identified as the knowledge management method, as workers from a past peak required "minimal briefing" the next year. The Airmee Site Manager mentioned weekly meetings of forecasting errors and analysis of operational deviations, and identified development of a dynamic, input-driven forecasting tool as the most important innovation, although the constant updating of this tool was considered to pose risks of systematic errors.

The Airmee Night Shift Team Leader evaluated the organizational learning culture in terms of achieved results: "Working there for one year has given me a clear picture of how this company has been able to grow – they have always made sure that they reflect on their past records. They have always made sure that they reflect on things that have gone wrong and the data they have, and to make sure that if there's anything that goes wrong, they take immediate measures – that's what they have been doing." He attributed the 6 percent yearly growth rate and 70 percent penetration into the Swedish population to this reflective practice.

The Bring Terminal Worker highlighted automation as the most significant innovation: "Bring has set up this DAVO sorting machine – it's really huge, and this machine is adjustable so it can deal with really high volumes. Then they had direct drop groups where some of the work is automated." He described his own innovations related to robotization: "We have these AMRs, autonomous mobile robots – we're trying to see if these can help in some of the processes, for example, can they drop some pallets, so in case we have a high spike in volumes, these robots can help in driving the pallets. Something we've been looking at, but

still no success, because the human is always more efficient – but it's something being looked at."

The PostNord Area Manager identified the organization's investments in harvesting its scan data as the most significant innovation. In addition, he mentioned the interest in future autonomous deliveries, considering drone delivery as possible in 3-5 years. The Bring Operations Coordinator identified the tool of real-time destination-level volume visibility as the most important missing element. The UPS respondent noted the lack of formal feedback from drivers preventing return of the route knowledge accumulated during daily work to the planning system. The Bonway distributor described no organizational learning mechanisms whatsoever, offering the operational provision of an assistance driver on high volume days as the most important improvement measure.

5. Analysis

This chapter provides an in-depth interpretation and analysis to answer the overarching research question: How does demand variability impact operational management and decision-making in parcel service providers? This analysis is derived from the empirical findings presented in Chapter 4 through the use of interviews which were conducted with logistics and operational management professionals from multiple companies such as Bring, Bonway, Airmee, PostNord and UPS.

In accordance with the methodological design, this analysis is structured into six themes by taking into account both the results section of this study and the interview guide. This helps to maintain internal consistency and draw comparisons between existing theoretical frameworks and empirical observations.

The analysis is based on the theoretical concepts of the Bullwhip Effect (Lee et al., 1997), Dynamic Capabilities (Teece, 2007), agile supply chains (Christopher & Towill, 2001) and capacity constraints (Slack et al., 2019).

Additionally, this analysis will carefully interpret the implications of the findings, identify effective and practical mechanisms, critically assess underlying modern logistical problems and cross examine how the existing theories explain the observed realities.

The findings highlighted the fact that demand volatility is a multifaceted phenomenon caused by predictable patterns and unexpected disruptions. The environment of logistics and delivery systems are highly dynamic and there are rapid changes and external factors that unexpectedly disrupt the operations and the delivery network. Therefore, to ensure operational management and the delivery system does not get disrupted, the parcel service providers and the management must continuously adapt to the complex and uncertain environment rather than depending on forecasting systems and historical data (Christopher, 2016).

This chapter's goal is to systematically link the empirical data with the theoretical concepts in order to comprehend how demand volatility is handled, experienced, managed and included within parcel logistics operations.

5.1 Experience of Demand Volatility and Its Drivers

The empirical evidence shows that parcel service providers do not view demand volatility as an unusual disruption, but instead as a consistent and inherent aspect of their operational and management landscape. These findings also closely correspond with existing research on last-mile delivery which suggests that demand volatility has emerged as a fundamental trait in contemporary supply chains and logistics due to the growth of e-commerce and the fragmentation of consumer demand (Savelsbergh & van Woensel, 2016; Pahwa & Jaller, 2023). Additionally, the findings also reveal that the operational actors view demand volatility not as an anomaly but rather as an expected aspect of regular daily operations.

The findings reveal that demand volatility can be both predictable and unpredictable depending on the time frame and operational and management circumstances. But firms like Airmee claim to achieve a fairly high forecast accuracy by reaching an accuracy rate of 95%, thereby stressing the importance that historical data and consistent trends and patterns such as seasonal peaks and mid-week demand, offers a more reliable foundation for planning. This supports the notion that demand trends in logistics and delivery systems frequently exhibit identifiable trends which allows forecasting models to be more accurate and effective (Simchi-Levi et al., 2008).

Through the data collected a key observation that was made is the dual characteristic nature of demand volatility which consists of both predictable and unpredictable factors. If an

aggregate analysis is conducted then the demand will show consistent trends and patterns. For example, it can be predicted that during seasonal peaks such as Christmas and Black Friday, the demand for products will increase and the volume of parcels to be delivered during these seasons will be very high. Therefore, during these seasons and recurring patterns, the problems caused by demand volatility can be solved through the use of historical data and forecasts.

Another example that was discovered through aggregate analysis is that B2B-oriented operations encounter very high demand at the beginning of the week. On the contrary, B2C-oriented operations encounter very high demand during mid-week. These examples highlight the fact that when demand is predictable and when there are recurring trends and patterns, the issue of demand volatility can effectively be resolved through the use of historical data and forecasts (Simchi-Levi et al., 2008).

However, the findings of this study reveal that even if the demand is predictable the operational reality remains highly unpredictable and uncertain. All the respondents of this study stated that forecasts are inaccurate because when there is an increase in the volume of packages the actual number of packages cannot be determined because there are other factors in play which are beyond organizational control. These factors include upstream delays, transportation disruptions, power failure at other terminals, retailer production problems, traffic and adverse weather conditions. Additionally, one respondent shared that in the event of unplanned additional shipments, the forecasts fail to take into account this additional number of parcels and these lead to poor decision making thereby disrupting the entire delivery network.

This phenomenon can be analyzed through the theoretical concept of the Bullwhip Effect (Lee et al., 1997). By applying the Bullwhip Effect, it is clear that even a minor disruption to the logistics and delivery network such as an inaccurate retailer forecast or even a delayed shipment can have a significant effect on the operational management. This means that a single disruption can put the operational processes which include sorting, routing and employing workers into chaos thereby resulting in the entire delivery network to fail.

Furthermore, the findings reveal that the Bullwhip Effect is very beneficial as it operates at a faster pace. This means that the Bullwhip Effect comprehends that at any time the demand and volume of packages can increase or decrease. Therefore, wasting time on planning cycles

is unnecessary. Instead, more focus should be to quickly adjust the logistics and delivery system to these rapid changes. This also highlights the fact that not all traditional supply chain theories are applicable to highly dynamic and time-sensitive environments such as parcel delivery and logistics.

In addition, the interviewees stated that consistent events like Black Friday, Christmas and seasonal sales can increase the volume of packages. This indicates that even economic conditions, promotional campaigns and consumer behavior play a central role in shaping demand trends and patterns (Christopher, 2016).

Another vital observation made through the Contingency Theory (Donaldson, 2001) is that demand volatility is shaped by organizational factors such as logistics structures, operational structures, service models, customer segmentation and the geographical scope of the operations and delivery network. The Contingency Theory emphasizes that every company does not follow the same universal principles because each company has their own operational processes and therefore is very distinct from one another. Therefore, to overcome the problems of demand volatility it's crucial to identify the nature of the company and its organizational factors.

The findings also can be interpreted to view demand volatility as a multi-layered phenomenon because at the macro level there are predictable patterns and trends but at the micro level there are unpredictable deviations.

It is crucial for the operational management to view demand volatility as a multi-layered phenomenon because when there are predictable patterns, they have to undertake forecasts and planning cycles and when there are unpredictable disruptions the parcel service providers must be prepared to adapt and quickly adjust the delivery system.

For example, at Bring the managers complained that the number of daily volumes of packages cannot be properly ascertained and the forecasts are inaccurate which are caused by the lack of transparency in the upstream operations and the unpredictable purchasing power of the consumers.

Managers of Airmee also complained that transit delays and backlog accumulation are the key sources of demand variability. This explains that even when demand is predictable at the

macro level there will still be uncertainty due to the decentralized nature of the logistics and delivery networks.

Therefore, these findings reflect that relying on historical data and forecasting is a waste of money and effort because it fails to capture real-time variability. Instead parcel delivery operators must use adaptive operational strategies which are far more effective in dealing with volatility and uncertainty.

5.2 Operational Challenges

The empirical findings highlight the fact that when the demand is volatile it is the parcel logistics systems and structures that are most affected and this gives rise to many operational challenges. The entire parcel logistics system will be badly affected because these operational structures and resources are highly rigid and cannot be easily adjusted or changed.

Sorting operations and facilities during a demand volatile situation will be affected very negatively. The findings show that during peak periods, there will be capacity constraints and the sorting facilities have to process large volumes of parcels within narrow time windows. Undertaking such a complex task will lead to bottlenecks that cannot be easily alleviated. Such massive volumes of packages can also exhaust the machinery, increase the electricity usage and occupy available space. The operations management theory stated that bottlenecks are a major disadvantage to the entire logistics delivery system which will lead to poor delivery system performance (Slack et al., 2019).

These problems at the sorting facilities cannot be easily adjustable in the short term because unlike labor these infrastructure, machinery and equipment are fixed resources that cannot be changed when the volumes of packages increase. Therefore, companies rely on reactive strategies such as redistributing packages to their other facilities, prioritizing certain deliveries over others and delaying the processing of certain packages. This demonstrates the weaknesses of the entire logistics structure which was built only to increase efficiency but during demand volatility the structure has failed to provide flexibility and adaptability.

Workforce management is another component of the logistics structure that will be negatively affected during fluctuating demand periods. When the volumes of packages increase there will be a massive workload, lack of available staff, lack of space for the workers to operate,

time pressure to sort out all the packages and there will be increased mental strain and fatigue on the employees. On the contrary, when there are less volumes of packages there will be cost inefficiency, less productivity and underutilization of the staff. This demonstrates how poorly the supply chain management is structured as it fails to be flexible and efficient (Christopher & Towill, 2001).

Human and psychological dimensions in logistics and operational management is one of the poorly researched areas in logistics literature. In the situation of increased volumes of parcels, the managers who are part of the operational management will be heavily under stress and they will be under enormous pressure because in order to adapt and make the necessary changes to the increased demand, the managers have to make quick decisions. Even the employees especially those in the sorting facilities will face physical and cognitive strain and fatigue which can affect performance and service quality.

Furthermore, during the increase in the volumes of packages the transportation and the delivery network coordination will be heavily disrupted. When there are more packages to be delivered the company must have available vehicles, drivers and external carriers. The findings show that many of the companies often struggle with securing additional drivers and vehicles and even external carriers will not be very helpful if they are also running on full capacity in such a crucial situation. Regulations, contractual obligations and sustainable requirements will be another problem when trying to outsource operations and resources.

This reflects the interdependency in logistics networks because if one part of the system is disrupted there will be a chain reaction and the entire delivery network will fail (Simchi-Levi et al., 2008). This is further proven by the empirical data available that emphasizes that failure in one terminal can affect the entire delivery network and the negative impact can extend beyond a single day.

Another important insight is that above we discussed that during increased volumes of parcels there was less machinery which creates bottlenecks. On the other hand, when there are less volumes of parcels the machinery will be under-utilized and result in inefficiency. So, they are structurally interconnected and they provide limited adaptability. The problem does not lie with insufficient capacity but on why these resources are way too rigid, why they lack flexibility and fail to adapt during difficult times.

In summary, demand volatility exposes the drawbacks of the fundamental characteristics of logistics and delivery systems and companies must improve the flexibility and adaptability in their operational structures and when allocating their resources.

5.3 Managerial Responses and Adjustment Strategies

To overcome the challenges caused by demand volatility, companies adopt a variety of managerial strategies that aid to improve organizational flexibility, adaptability and to ensure that the delivery network system performs efficiently and effectively.

In the findings it is revealed that the most significant managerial strategy is the use of flexible labour arrangements. This means that the company must recruit a mix of permanent employees, temporary employees and on-demand employees. This strategy is highly effective when the demand is volatile because even if there is an increase or decrease in the volumes of parcels, the managers can simply adjust the number of employees needed to complete the tasks which makes the workforce very flexible and efficient (Auad et al., 2026).

Through such a recruitment strategy the company can build a flexible labour pool and with time and experience such a mixture of different types of employees will be well trained and experienced to handle any situation even worse than demand volatility. This strategy even helps to reduce excessive costs which can help increase profitability.

But like any other strategy, flexibility in labour has its limitations. Having a mixture of different types of employees can be very difficult to train and coordinate. Especially, temporary employees because they lack the training and will not be familiar with the sorting facilities and how to operate the machinery which can lead to inefficiencies. Therefore, this strategy can reduce excessive costs but there will be operational instability and inefficiency.

In addition, another strategy is to prioritize specific shipments over others. This means that when the demand is volatile and there is an increase in the volumes of packages, the managers must instruct the employees to categorize the packages based on their delivery date. The time sensitive deliveries must be prioritized and must be quickly delivered to consumers as it helps to improve customer satisfaction and customer retention (Christopher, 2016).

The third managerial strategy that can be used is dynamic route and traffic management. This means that if there is traffic, congestion or adverse weather conditions, the managers must

decide to change the delivery routes, redistribute the packages to another facility and based on real-time information they must reallocate the resources. Empirical findings highlighted that for the logistics and delivery system to operate efficiently without any disruptions, it is mandatory for the managers to have a fully functioning dynamic route and traffic management system and there must also be a team who is well experienced and knowledgeable to handle and monitor such a system. Dynamic vehicle routing is also a similar strategy that helps to improve the last-mile delivery system (Liu & Wang, 2026).

The fourth managerial strategy is to collaborate with subcontractors and third-party carriers thereby expanding external capacity. This strategy is very similar to the outsourcing strategies theory which is used in supply chain management (Zhu et al., 2025). In the event of increased volumes of packages, the company does not need to waste time and money in purchasing more vehicles and more drivers. Instead, the company can rely on a third-party carrier which can help to deliver the packages on time and reduce excessive costs. But relying on external partners can reduce service quality, reduce control and there will be dependency on sub-contractor availability.

Through the discussions of these four managerial strategies, a critical insight that can be identified is that these four strategies do not solve the problem of demand volatility, instead they shift and redistribute its effects. This means that, forming a flexible labour pool can reduce excessive costs but will be very difficult to coordinate and manage, an effective dynamic route and traffic management system can ensure the parcels are delivered on time but places high cognitive pressure on the managers to make quick decisions and collaborating with an external party can reduce excessive costs on transportation and labour but introduces financial and service quality risks.

Therefore, demand volatility cannot be fully eliminated through only these 4 managerial strategies. And these 4 strategies will only be effective if they are carefully designed and implemented by the managers and they also depend on the managers' ability to make decisions in situations involving time pressure and uncertainty.

5.4 Decision-Making Under Uncertainty

The empirical findings reveal that the managers who operate in parcel delivery operations make their decisions based on time pressure, uncertainty and lack of information. Managers

who have experience in the field of logistics and parcel delivery operations comprehend that the working environment is highly subject to change and adaptability. Therefore, managers don't make their decisions purely on forecasts and data, instead they rely on analytical tools and human judgement.

When conducting the interviews most of the respondents stated that they relied on a "gut feeling" or "fingertip feeling" before they made their decisions which demonstrates that experience and intuition are two significant skills that managers must develop and harness. Such an observation can be interpreted through the dual-process theory. The dual-process theory states there is a difference in quick intuitive decision making and slow analytical decision making (Kahneman, 2011).

Managers within the logistics operations have to make quick decisions and they have to rely on their experiences and intuition because in such an environment the managers don't have the luxury of waiting for the completion of forecasts and detailed analysis. For instance, the moment the Bring managers receive any new information, they make the staffing decisions and assign their specific roles within the hour of receiving the new information. And PostNord managers make their decisions based on years of experience in comprehending demand patterns and trends. Therefore, these findings reveal that the gap between planning and execution is closed and filled through experience and intuition.

However, sole reliance on intuition and experience can cause problems because intuition is based on cognitive biases, personal assumptions or incomplete information. Furthermore, the findings state that data cannot be replaced with intuition, instead intuition is only a complement to the analytical tools. Analytical tools are built through the combination of multiple sources of information such as through historical trends and patterns, forecasted volumes of parcels, real-time communication and knowledge and experiences of the operations. Such a hybrid decision-making process that reflects the complexity of the logistics environment can help make rational decisions as they are not solely based on data or experience. Therefore, companies must have a balance between allowing managers the flexibility to act promptly and ensuring that the decisions are supported by reliable data.

In addition, in a situation of uncertainty a collective process involving multiple actors such as drivers, coordinators and managers across the different levels of the company can help to make better decisions. Such a decentralized decision-making structure distributes the

authority to make the decision amongst these 3 actors is one of the best strategies to deal with local conditions and problems.

In summary, during a demand volatile crisis the management must make decisions based on experience, adaptability and promptness. Analytical tools are helpful in other industries but not in the logistics and operational management industry because analytical tools don't fully capture real-world operations and its complexities. Therefore, human judgement remains the most important element for effective decision making in parcel logistics and delivery systems.

5.5 Planning-Level vs Execution-Level Perspectives

The findings reveal that there is a systematic gap between planning and execution. Planning processes are formed through forecasts, structured models and predefined assumptions but they deviate from the two key characteristics of logistic operations which are real-time disruptions and changing conditions. For example, in order to accurately forecast the volume of parcels, Bring used an AI-based forecasting system. But the forecast was inaccurate as there were deviations of up to 10,000 parcels in a single day.

Furthermore, the findings show that in the environment of parcel logistics if the level of operational details increase then there is a high probability that the planned forecast will be inaccurate. This means that long-term forecasts can be accurate but short-term forecasts result in continuous errors because in the environment of parcel logistics there are continuous changes and disruptions that are not taken into account by these forecasts (Simchi-Levi et al., 2008).

To minimize the errors in the forecasts, managers combined the forecasts with real-time data and experience-based judgement. For example, when conducting the interviews most of the respondents stated their forecast was combined with the latest trends and patterns, communication information with the other departments and personal experience. Therefore, such forecasting will be broadening the adaptive process rather than being used to make decisions.

In addition, at Bring the managers complained that their forecast was inaccurate as there were large deviations in daily volumes because unpredictable customer behaviors and the lack of

visibility into upstream operations were not taken into account. This means that planners ignore the variability of incoming volumes and only include aggregated data and forecasts. Therefore, the primary cause of this gap is called information asymmetry.

Airmee also highlighted the fact that their forecast was inaccurate because it failed to take into account the variability caused by the transit disruptions and backlog accumulation.

Operational disruptions such as delays in the delivery vehicle, unexpected increases in volume, system and equipment failure cannot be anticipated so they were ignored from the planning phase which created a gap between planning and execution. These disruptions are not special events but they occur regularly so there is a reinforced need for continuous adaptation. Therefore, execution becomes a dynamic process which continuously evolves.

The planning process in parcel logistics will function as a probabilistic framework which will provide general guideless not precise instructions. Therefore, the concept of dynamic capabilities will be highly relevant because in response to the changing conditions the company must have the ability to reconfigure its resources and processes to match these changing conditions (Teece, 2007).

Even if the company improves the accuracy of the planned forecasts this will not close the gap between planning and execution because a forecast cannot take into account all the possible disruptions and variations. So, in order to close the gap between planning and execution, the company must focus on improving both planning and adapting to the execution stage.

Additionally, frontline staff such as drivers, coordinators and terminal workers can bridge the gap between planning and execution because these employees possess local knowledge and real-time experience and insights that are not available at the planning stage.

In summary, the gap between planning and execution cannot be fully resolved because this gap needs to be managed. For there to be effective operations there must be both robust planning systems and flexible and adaptable execution processes along with competent communication mechanisms that help the company to deal with certainty in real time.

5.6 Organizational Learning and Future Adaptations

The empirical findings state that parcel service providers are highly interested in organizational learning because it can help them to overcome the problems of demand volatility. Organizational learning is being done through both formal mechanisms such as data analysis and reviews and informal mechanisms such as by sharing your knowledge with your employees and gaining or accumulating more experience.

Organizational learning is amplified through post-event analysis. Post-event analysis means that when the demand increases and the number of volumes of parcel increases, the company will record and document their actions, outcomes of what they did during this demand volatile period and they will also identify the areas that can be improved. Through post-event analysis the company employees can learn from the past experiences and reflect and this can even help to make future decisions (Argyris & Schön, 1978). Post-event analysis can also analyze past performances and determine how to refine their strategies and improve their preparedness for future obstacles.

Additionally, Tacit knowledge must be introduced and this is where managers who have worked in logistics for many years share their excellent knowledge and experience in handling demand volatile situations.

Furthermore, organizational learning is improved by adopting data analytics, Artificial intelligence and automation which can result in highly accurate forecasts and will help to make the operations and employees more efficient and productive. Scan data and machine learning models are also being used to get a detailed analysis of parcel flows and demand patterns capabilities (Wamba et al., 2017).

In summary, organizational learning along with technological advancements can improve their ability to deal with demand volatility over time.

5.7 Synthesis of Findings

This detailed analysis of the six themes demonstrates that demand volatility is a systemic and structural condition that negatively affects the entire parcel logistics company. The findings also reveal that certain demand patterns can be predicted but the issue lies within the operational environment which remains uncertain because there can be external disruptions,

lack of information and also due to the interconnected nature of parcel logistics and delivery networks.

A key insight is that the problems caused by demand volatility cannot be solved by a single strategy. Parcel logistics companies must adopt a multi-dimensional strategy which combines forecasting, real-time adaptability and flexibility to overcome challenges posed by demand volatility. For this multi-dimensional strategy to be successful the company must balance its competing priorities.

The analysis also reveals that there can be advancements in technology and data analytics but the ultimate decision-making responsibility and power falls to humans and their decisions will be based on their experiences, intuition and communication. This also emphasizes the fact that technological solutions should not replace humans but instead should be designed to support humans.

Additionally, the analysis highlights that through organizational learning, the company can derive lessons and wisdom from their past experiences, they can use these past experiences to improve their processes, they can formulate more flexible structures that can be well equipped to handle and overcome demand volatility and uncertainty. Therefore, managing volatility is an ongoing process of flexibility, adaptability and education not just a one-time effort.

5.8 Conclusion of Analysis

In conclusion, the analysis highlighted the fact that demand volatility has a negative impact on the decision-making process and every aspect and department of the parcel logistics company. Demand volatility is sourced from the very structural characteristics of the logistics and delivery systems and this is the very reason why it cannot be eliminated but this analysis proves that managerial strategies can help to overcome demand volatility.

The findings also demonstrated that demand volatility can be effectively managed through the combination of organizational learning, flexible resources and adaptive decision making. Also, companies should not rely on technology but must invest on human capital and design technology to support humans and prioritize human judgment.

6 Conclusions

This chapter outlines the conclusions drawn from the research based directly on its research questions. Starting from the overarching research question of how demand variability impacts operational management and decision-making in parcel service providers, four conclusions are presented, each related to one of the specific sub-questions informing the empirical investigation.

Fluctuations in parcel volume: What impact do they have on route planning, stop density, and resource utilization throughout various operational shifts?

The most reasonable way to interpret the evidence is that parcel volume fluctuations impact route planning, stop density, and resource utilization in a way that is structurally patterned and predictable but practically disruptive in implementation. There is a systematic difference in the weekly demand curve by type of operation: companies oriented towards business-to-business operations experience the greatest demand at the beginning of the week, whereas companies operating on the business-to-consumer logic see peak demand mid-week. In both cases, seasonal peaks, particularly the Christmas peak, multiply the effect by anywhere between two and three times the baseline demand (in the evidence presented), with extreme cases multiplying demand up to three hundred percent compared to the base level. In either case, the practical implication is that there is an inevitability that leads to deviating from the planned route: load redistribution among other drivers, increasing density in existing clusters, and, ultimately, sending excess cargo back to the terminal. Capacity, equipment, and personnel operate in the exact same way; peak periods see acute pressure on these resources while the troughs see them idle.

High-demand periods vs low-demand periods: What operational challenges do they pose and what are the differences?

When considering the evidence, it becomes clear that these two periods represent not just challenges of different magnitude but entirely different natures as well. In the case of the former, the challenge is the presence of a capacity ceiling that cannot be adjusted fast enough to accommodate the surge without sacrificing quality or paying excessively to do so. This ceiling is composed of multiple aspects: throughput limitations in sorting operations, the

number of carriers used in contract negotiations, and personnel endurance all appear as critical constraints. On the other hand, the trough poses the opposite challenge: excessive idle capacity in the form of expensive and fixed equipment, labor costs, and associated expenses. The most reasonable way to interpret these findings is that the two problems are structurally related: capacity rigidity gives rise to overload at peak periods and underutilization at troughs, thus making solving one problem exacerbate the other.

What strategies do managers and coordinators use to tackle demand variability?

The evidence suggests that the most important response to demand variability is achieved via flexible employment arrangements, route and traffic adjustments, external capacity management, and organizational culture. Flexible employment schemes are clearly the most frequently mentioned strategy; they cover everything from simple on-demand pools to hiring via agencies and using student labor. The second most prevalent mechanism is adjusting routes and loads, particularly in cases of splitting the load or reallocating it between trucks and even cancelling certain trips. External carrier management offers some additional flexibility, limited in some cases by contractual obligations and sustainable carrier practices in another case. Finally, perhaps counterintuitively, organizational culture appeared as a management tool in one of the organizations, bringing a cultural aspect to the issue that is more important than initially assumed by current literature.

Do the strategies described alleviate the strain, or do they create additional inefficiencies?

The evidence suggests that the strategies listed above partially alleviate the strain but fail to address it entirely, creating secondary inefficiencies in their wake. Flexible labor reduces cost exposure but imposes a significant coordination overhead and may present a risk of poor quality when pushed to its extremes. Standing standby agreements with carriers help limit the effect of surge periods on costs but create additional costs at lower periods. Splitting and redistributing load is a powerful strategy for absorbing extra cargo but adds stress on coordinators and may produce a cascade of problems depending on working time and circular tour restrictions. The planning-execution gap, observed throughout all five companies, was due not to forecasting errors but rather to systemic issues that are inherent to any closed informational system and variable upstream processes and data limitations.

7 Discussion

7.1 Beyond the Empirical Findings

The data analysis yields insights that extend beyond what can be definitively verified by the interview data itself, yet which the authors feel should be considered thought-provoking conjectures about the nature of the problem.

First among these insights was the extent to which experiential judgment gut feeling, fingertip feeling, intuition was repeatedly cited as not being a fallback in lieu of superior analytical tools, but rather as an active component of the decision-making process, conducted concurrently and independently of the algorithmic forecast and not subservient to it. This was observed even among firms like PostNord and Bring, which used highly accurate forecasting algorithms or prognoses; indeed, the managers who most positively assessed these technologies were also the ones who understood their limitations best and who most firmly believed that their own judgment added value to the equation. This suggests that while it is common in the literature on automation to assume that there will be a gradual substitution of experiential judgment with technological tools, in the context of demand volatility management, the dynamic might in fact involve ever-greater interdependence between human experience and algorithmic forecasting, such that the former's interpretative burden becomes heavier rather than lighter as the latter matures.

Secondly, there is the question of organizational culture in the context of operational resilience. The fact that Bring made explicit reference to its culture as an integral part of its strategy, as an organizational disposition rather than as an outcome of resource investment, is significant for the way in which it describes resilience as something achieved not through structural means, but rather through managerial behavior and communication. The ability of a firm to cultivate such a disposition in its employees, and whether and under what conditions this ability breaks down, is a question the current data does not answer, but which appears worthy of exploration.

7.2 Implications for Practitioners

Several practical implications can be drawn from the data with respect to practitioners and business managers working within the parcel logistics sector. First, the evidence highlights

the importance of flexible labor arrangement as a deliberate and conscious management strategy - specifically in the sense that having access to a labor pool was not sufficient to ensure rapid response to disruptions. Organizations with well-established, trusted connections to a labor pool seemed better prepared to manage disruptions. These relationships need to be actively built and maintained, which cannot be done simply by employing people on the books, nor in terms of cost per hour.

Second, there is reason to believe that information architecture is a critical element of operational resilience, as illustrated by the Airmee Site Manager's observation that closed-loop communication was the biggest obstacle to closing the planning-execution gap; in other organizations, the effects of the disruption had extended farther than necessary precisely due to the failure to communicate with all relevant parties. While there was benefit to having relevant information broadcast on an open channel, even when not everyone in the loop was directly implicated, this also has implications for how communication must be designed internally as well as contractually between the parcel operator and its retailer clients.

Thirdly, it is worth considering the risk posed by obsolete operational technology, exemplified by the mobile data terminal described in UPS; despite its inefficiencies, it was identified by the respondent as being a greater source of possible improvements to operational efficiency than route optimization or staff changes. When operational technology is assumed to be a fixed cost, this discrepancy may go unnoticed.

7.3 Implications for Future Research

A number of avenues for further investigation are suggested by this study. The first relates to generalization of the current findings across other classes of parcel operators. All five companies included in the sample are relatively large, well-established companies with their own management structure, dedicated terminal infrastructure, and labor arrangements. Operators of smaller scale, new entrants to the field, including gig economy-based last-mile solutions, are likely to manage demand volatility via different channels and have different adjustment possibilities. Cross-comparison of different operators with respect to volatility management would greatly expand our understanding of the subject matter.

A second avenue concerns the longitudinal dimension of planning-execution mismatch that

the current study does not consider. In the words of the Bring Production Manager, companies develop volatility management capabilities over time: learning how to plan the work with twelve thousand parcels per day versus one hundred thousand, improving year by year based on a review of previous peak periods. Longitudinal case study research focused on the changing nature of planning-execution mismatch over time, as companies grow and accumulate operational experience, and acquire new technologies, would provide valuable additional insights to the current literature.

A third avenue of investigation relates to the relationship between the carriers and retailers upstream of the operators, and specifically, the inability of operators to get information about the destination distribution of incoming traffic. As noted by the Bring Operations Coordinator, the inability to control the data about where the traffic was headed was by far the most relevant data constraint for her in planning. Further investigation into the structures that created such information asymmetry and conditions for overcoming it would provide a useful practical contribution.

7.4 Methodological Reflection

Strengths of the current study include its detailed empirical base and the multi-organizational and multi-role design that allows interorganizational comparative research at both levels of the planning-execution gap. The use of semi-structured interviews allowed the respondents to bring up themes of a non-intentional kind, especially the psychological aspects of peak management that had not been anticipated during preparation of the interview guide yet turned out analytically relevant. Having both managerial and front-line perspectives in one sample provided a richer understanding of planning-execution mismatch than any one of these perspectives would have alone.

One obvious weakness of the sample is its small size. Eight interviews across five operators provided sufficient data for reaching a high degree of thematic saturation and generating substantial analytical insights; yet statistically valid representation cannot be claimed in light of such sample sizes. Thus, the conclusions should be regarded as analytically generalizable rather than universally applicable. The geographical context of the research, namely, Sweden, also brings about a contextualized component of results, since the specificities of the country's labor market, sustainable development requirements for carrier relationships, and public holiday schedule are peculiar to it.

The possibility of researcher influence on the interview process should also be considered. It must be understood that semi-structured interviews are by definition jointly constructed, and the questions posed by the researcher set up the discussion's agenda. The emergence of unexpected themes, such as economics of the service ceiling, or specifics of the carrier standby, depended on the particular interview, even though they could potentially apply to all the companies involved. More comprehensive methods of data gathering, such as observation and documents analysis, might help to explore this angle in future research.

All things considered, the study gives a richly empirical account of the problem of demand volatility at the operational level of parcel carriers, which receives insufficient attention in the current literature; in doing so, it lays a foundation for further research.

References

Argyris, C., & Schön, D. A. (1978). *Organizational learning: A theory of action perspective.* Addison-Wesley.

Argyropoulou, M., et al. (2023). Horizontal collaboration in the last mile distribution: Gauging managerial response to disruption and abnormal demand. *Benchmarking: An International Journal, 30*(2), 460–474. https://doi.org/10.1108/BIJ-06-2021-0328

Auad, R., Fillebeen, T., Levkin, R., Rakshit, A., & Savelsbergh, M. (2026). Balancing flex and non-flex labor to reliably meet on-demand capacity. *Transportation Research Part E: Logistics and Transportation Review, 209*, Article 104696. https://doi.org/10.1016/j.tre.2026.104696

Aylak, U. (2025). Large language models in supply chain management: A systematic review. *International Journal of Production Research.* Advance online publication. https://doi.org/10.1080/00207543.2026.2641103

Boyer, K. K., & Hult, G. T. M. (2005). Customer behavior in an online ordering application: A decision scoring model. *Decision Sciences, 36*(4), 569–598. https://doi.org/10.1111/j.1540-5414.2005.00103.x

Braun, V., & Clarke, V. (2006). *Using thematic analysis in psychology*. *Qualitative Research in Psychology, 3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Campbell, J. F., & O'Kelly, M. E. (2012). Twenty-five years of hub location research. *European Journal of Operational Research, 222*(2), 153–169. https://doi.org/10.1016/j.ejor.2011.10.024

Christopher, M. (2016). *Logistics & supply chain management* (5th ed.). Pearson.

Christopher, M., & Towill, D. R. (2001). An integrated model for the design of agile supply chains. *International Journal of Physical Distribution & Logistics Management, 31*(4), 235–246. https://doi.org/10.1108/09600030110394914

Cooper, T., Reagan, I., Porter, C., & Precourt, C. (2019). *Global fleet & MRO market forecast commentary 2019–2029.* Oliver Wyman.

Donaldson, L. (2001). *The contingency theory of organizations.* Sage Publications. https://doi.org/10.4135/9781452229249

Galkin, A., Švadlenka, L., Vrba, R., & Kijewska, K. (2025). Navigating the future of urban logistics: Conceptual framework for dynamic freight management. *Transportation Research Part D: Transport and Environment, 147*, Article 104956. https://doi.org/10.1016/j.trd.2025.104956

Gevaers, R., Van de Voorde, E., & Vanelslander, T. (2011). Characteristics and typology of last-mile logistics from an innovation perspective in an urban context.

Kahneman, D. (2011). *Thinking, fast and slow.* Farrar, Straus and Giroux.

Lee, H. L., Padmanabhan, V., & Whang, S. (1997). Information distortion in a supply chain: The bullwhip effect. *Management Science, 43*(4), 546–558. https://doi.org/10.1287/mnsc.43.4.546

Leskinen, J. (2021). *Optimization of logistics service provider's parcel network* (Master's thesis, Aalto University).

Lim, S. F., et al. (2018). Last-mile logistics: A systematic review and typology. *Sustainability, 11*(24), Article 7131. https://doi.org/10.3390/su11247131

Liu, L., & Wang, T. (2026). Optimizing urban last-mile delivery efficiency through dynamic vehicle routing heuristics and traffic flow analysis. *Scientific Reports, 16*, Article 483. https://doi.org/10.1038/s41598-025-29861-y

Michalski, M., Montes-Botella, J., & Narasimhan, R. (2018). The impact of asymmetry on performance in different collaboration and integration environments in supply chain management. *Supply Chain Management: An International Journal, 23*(1), 33–49. https://doi.org/10.1108/SCM-09-2017-0283

Núñez-Merino, M., et al. (2025). Optimizing last-mile delivery through data-driven transformation: An action research approach. *Journal of Enterprise Information Management.* Advance online publication. https://doi.org/10.1108/JEIM-07-2025-0579

Pahwa, A., & Jaller, M. (2023). Assessing last-mile distribution resilience under demand disruptions. *Transportation Research Part E: Logistics and Transportation Review, 172*, Article 103066. https://doi.org/10.1016/j.tre.2023.103066

Savelsbergh, M. W. P., & van Woensel, T. (2016). City logistics: Challenges and opportunities. *Transportation Science, 50*(2), 579–590. https://doi.org/10.1287/trsc.2016.0675

Simchi-Levi, D., Kaminsky, P., & Simchi-Levi, E. (2008). *Designing and managing the supply chain: Concepts, strategies, and case studies* (3rd ed.). McGraw-Hill.

Slack, N., Brandon-Jones, A., & Johnston, R. (2019). *Operations management* (9th ed.). Pearson.

Suzuki, Y., Chen, H., & Faires, C. (2025). Response strategies for demand surges in last-mile

logistics: Managing delivery efficiency in volatile environments. *Journal of Business Logistics, 46*(1), Article e70004. https://doi.org/10.1111/jbl.70004

Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal, 28*(13), 1319–1350. https://doi.org/10.1002/smj.640

Wamba, S. F., Gunasekaran, A., Akter, S., Ren, S. J. F., Dubey, R., & Childe, S. J. (2017). Big data analytics and firm performance: Effects of dynamic capabilities. *Journal of Business Research, 70*, 356–365. https://doi.org/10.1016/j.jbusres.2016.08.009

World Economic Forum. (2025). *Transforming urban logistics: Sustainable and efficient last-mile delivery in cities.*

World Journal of Advanced Engineering Technology and Sciences. (2025). AI-driven forecasting and machine learning in modern supply chains. *World Journal of Advanced Engineering Technology and Sciences, 15*(2), 680–686.

Zhu, L., Adulyasak, Y., & Rousseau, L.-M. (2025). Partial-outsourcing strategy for the vehicle routing problem with stochastic demands. *Networks.* Advance online publication. https://doi.org/10.1002/net.70024

Appendix A: Interview Guide

Introduction for Participants and consent

Thank you for participating in this study. The purpose of this interview is to understand how demand volatility affects operational management and decision-making in parcel service providers. There are no right or wrong answers-we are interested in your experiences and perspectives. Your responses will be anonymous.

- Do we have your consent to proceed with the interview?
- Can you briefly describe your role at the organization?

Theme 1: Experience of Demand Volatility

1. How would you describe demand volatility in your region?
2. What are the main drivers of fluctuations in parcel volumes?
3. How predictable is demand from your perspective?
4. Can you describe a recent period of extreme volatility and how it affected operations?

Theme 2: Operational Challenges

5. What are the biggest operational challenges during high-demand periods?
6. How does volatility affect route density and resource allocation?
7. Do you see different challenges during low-demand periods?
8. How often do surges lead to delays or failed deliveries?

Theme 3: Managerial Responses and Adjustment Strategies

9. What strategies does the organization use to manage fluctuating demand?
10. How do you adjust staffing, routes, or schedules when demand changes suddenly?
11. Do you rely on temporary staff or subcontractors during peak periods?
12. What role does technology or forecasting tools play in your decision-making?
13. What trade-offs do you face between cost efficiency and service reliability?

Theme 4: Decision-Making Under Uncertainty

14. What information do you rely on when making decisions under uncertainty?
15. How quickly do you need to respond when demand shifts?

16. How does communication work between planners, terminals, and drivers during disruptions?

Theme 5: Planning-Level vs Execution-Level Perspectives

17. How well do planning-level forecasts match daily operational reality?
18. Where do you see the biggest gaps between planning and execution?
19. How do frontline staff provide feedback to planners?

Theme 6: Organizational Learning and Future Adaptation

20. How does the organization learn from past peak periods?
21. What improvements or innovations are needed to handle demand volatility more effectively in the future?

DECLARATION

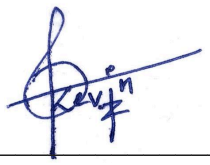
We hereby declare that this thesis represents our own work.

We declare that we have obtained the necessary ethical approval (e.g., GDPR) and acknowledged our obligations and the rights of the respondents in the research.

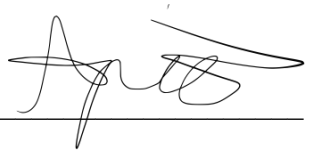
We declare that if we have made use of non-sensitive personal data we have collected GDPR Thesis consent forms, including contact details with participants in our study.

We have read and applied the Jönköping University's current research ethics guidelines with regards to the use of artificial intelligence (AI) tools in this work as presented in the Course Guide. During the preparation of this work the authors used the following AI tools and the purpose for their use. We used Claude for transcribing and summarizing the conducted interviews, Scopus AI for secondary research and Quillbot to review correct grammar throughout the work.

As authors, we have reviewed and edited the content as needed and take full responsibility for the content of the thesis.

Signature: 

Date: 22.05.2026

Signature: 

Date: 22.05.2026